

Fundamentals · Gym · Q-learning

95% of AI interviews ask about RL. 5% of AI jobs use it. Today we teach the vocabulary (for interviews), one real project (for understanding), and honest perspective on when to reach for it.

Name one RL system you interact with weekly. (Hint: look at Ola's dynamic pricing.)

WHAT'S INSIDE

- 01** Five terms — memorise
- 02** Explore vs exploit — the restaurant...
- 03** Q-learning in 30 lines
- 04** Model-free vs model-based
- 05** Where RL actually ships (Indian...

WHY THIS LESSON MATTERS

RL won Go. RL set Ola surge pricing. Where did the middle go?

NUMBER 1

95%

From real-world deployments.

NUMBER 2

5%

From real-world deployments.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

State, action, reward, policy, value.

2

Explore vs exploit trade-off.

3

Train a Q-learning agent.

4

Model-based vs model-free.

5

When RL beats supervised.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

**Five terms —
memorise**

State · Action · Reward · Policy · Value

02

**Explore vs exploit —
the...**

ϵ -greedy is the workhorse

03

**Q-learning in 30
lines**

Solve Frozen Lake

04

**Model-free vs model-
based**

Key concept of this lesson.

01

PART 1 OF 4

Five terms — memorise

State · Action · Reward · Policy · Value

CONCEPT 1 OF 4

Five terms — memorise

State (s) — current snapshot of environment.

Action (a) — what the agent can do.

Reward (r) — numeric signal to maximise.

Policy (π) — state $\rightarrow$ action mapping.

Value (V / Q) — expected cumulative reward.

VOCAB

5 terms

Every RL paper uses these exact symbols.

DEFINITION

Five terms — memorise

State · Action · Reward · Policy · Value

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Explore vs exploit — the restaurant dilemma

Exploit: keep visiting your favourite place.

Explore: try somewhere new.

ϵ -greedy: random action with probability ϵ .

Decay ϵ over training.

UCB / Thompson Sampling are fancier alternatives.

BALANCE

ϵ -greedy

Simple. Works. Use first.

KEY POINTS

- Exploit: keep visiting your favourite place
- Explore: try somewhere new
- ϵ -greedy: random action with probability ϵ
- Decay ϵ over training

CONCEPT 3 OF 4

Q-learning in 30 lines

Q[state, action] table.

Update: $Q[s,a] += \alpha \cdot (r + \gamma \cdot \max Q[s',\cdot] - Q[s,a])$

Run 2000 episodes; watch Q converge.

np.argmax(Q) gives the learned policy.

BELLMAN

$r + \gamma \cdot \max Q$

The equation behind all value-based RL.

DEFINITION

Q-learning in 30 lines

Solve Frozen Lake

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Model-free vs model-based

A core idea you'll use throughout this lesson.

KEY POINTS

- A core idea you'll use throughout this lesson
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Train on FrozenLake

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which problem in your life has a clear reward function?

Q2

Which doesn't — and why supervised would beat RL there?

Q3

When would you build a simulator before anything else?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Policy is:



Model



State → action mapping



Reward function



Value

Policy is:

CORRECT ANSWER



State → action mapping

WHY

$\pi: s \rightarrow a$

Q-learning is:

A

Model-based

B

Model-free

C

Supervised

D

Random

Q-learning is:

CORRECT ANSWER



Model-free

WHY

Learns Q-values directly from experience

$\epsilon = 0.1$ means:

A

Always greedy

B

10% random

C

Always random

D

Never random

$\epsilon = 0.1$ means:



CORRECT ANSWER

10% random

WHY

10% random action; 90% greedy

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

State (s) — current snapshot of environment.

Action (a) — what the agent can do.

Reward (r) — numeric signal to maximise.

Policy (π) — state $\rightarrow$ action mapping.

Value (V / Q) — expected cumulative reward.

VOCAB

5 terms

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THE TAKEAWAY

"RL is the interview favourite. Supervised is the job favourite."

WHAT TO NOTICE

1

Five terms — memorise

2

Explore vs exploit — the...

3

Q-learning in 30 lines

4

Model-free vs model-based

CartPole in Gym

Train Q-learning on CartPole.

DELIVERABLES

- pip install gymnasium.
- Implement ϵ -greedy Q-learning.
- Plot reward vs episode.
- Submit plot.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Reward curve image.

ONE THING TO REMEMBER

“

"RL is the interview favourite. Supervised is the job favourite."

— AI Learning Hub

END OF LESSON 7

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 4 — RLHF.